

Yousef Alsaqa

Mechatronics & Robotics Engineer, building production software and hardware systems

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EDUCATION

Queen's University, Kingston, ON, B.A.Sc., Mechatronics & Robotics Engineering

Sep 2021 – Jun 2025

Fourth-Year Dean's List (2025) · Principal's Scholarship for Academic Excellence (\$4,000)

WORK EXPERIENCE

Queen's University, Undergraduate Admission & Recruitment

Business Analyst

Jul 2026 – Present

- Finalized the document attacher into the office's main way of attaching documents, root-causing a redundant page-reload bug that was silently doubling its runtime. Now used office-wide, processing 20,000 to 30,000 documents in a single admission cycle.
- Automated a CEGEP curriculum evaluator and a high-school-average-to-student matcher, and grew the evaluation dashboard (started as an IB-only tool) into a ~5,800-line system driving 5 parallel PeopleSoft browser sessions across multiple curricula.
- Turned the Commerce-only admission analysis dashboard, started during the Recruitment Representative role once the by-hand analysis got too slow, into a fully generic, config-driven dashboard where a new program is a config file, not new code.

Admission Coordinator

Jan 2026 – Jun 2026

- Built the document attacher and a Tkinter/Selenium tool that semi-automated the admission-decision workflow (read the documents, type the average, the tool handles the rest), processing about 1,200 applications, and started an evaluation dashboard limited to IB students.

Recruitment Representative

May 2025 – Jan 2026

- Traveled across Ontario presenting to prospective students almost daily.
- Built an automatic, fuzzy-matched location analyzer (5 years, ~170K applications, feeder-school and regional conversion analysis) and ran the Commerce-side Random Forest analysis against first-year GPA. The negative-correlation finding on the supplementary application helped push admission thresholds up, and started the dashboard framework the Engineering-side analysis was later built into.

Mechanical Engineering Intern, Dorsch Holding GmbH, Abu Dhabi, UAE

Jun 2023 – Aug 2023

- Redesigned a water drainage system from pumped to gravity-fed on an oil-field expansion project after fluid simulations revealed inefficiencies, and reworked pipe network layouts to reduce pressure losses.

PROJECTS

WorldCupFantasy, full-stack fantasy football platform, live for the entire World Cup 2026

Jan 2026 – Present

- Solo-built a Next.js/TypeScript/PostgreSQL platform serving real users through the actual tournament; built a live-scoring engine reconciled in real time against an error-prone external data feed, verified with 3 hand-rolled test harnesses (1,600+ lines) and against a genuinely live match before trusting it on tournament data.
- Every admin action (point overrides, retroactive corrections) writes to an immutable, revertible audit log; forked the architecture into a second live product (a La Liga fantasy app) to prove the design wasn't tournament-specific.

CAST Medical Interface, RFID-based patient simulator for nursing education

Sep 2024 – May 2025

- Assistant lead for software and hardware integration on a senior capstone: 12 RFID-tagged anatomical positions, 28 audio scenarios, and a Flask interface on a Raspberry Pi for ~\$250 in hardware, deployed and in use at Queen's School of Nursing.

Queen's AutoDrive, Level 4 autonomous vehicle, SAE AutoDrive Challenge

Sep 2021 – May 2024

- Trained and optimized CNN-based lane-perception models for low-latency embedded inference (year 1); moved to system integration (year 2), synchronizing ROS2 sensor fusion, path planning, and control nodes over CAN bus at 10Hz.

Formula SAE Steering System, Electrical Lead, custom driver telemetry

Sep 2022 – May 2023

- Designed a custom PCB and TFT display delivering real-time telemetry over CAN bus at 500kbps; root-caused termination-resistor issues that were causing intermittent bus dropout at speed.

TECHNICAL SKILLS

Software & Web: TypeScript, Next.js/React, Node.js, Python, C/C++, SQL/PostgreSQL, SQLite, Prisma, JWT Auth, Git

Robotics & Embedded: ROS/ROS2, OpenCV, SLAM, CAN Bus, Arduino, Raspberry Pi, SolidWorks, MATLAB/Simulink

Data & ML: Pandas, Scikit-learn, PyTorch, Random Forest, rapidfuzz

Automation & Tools: Selenium, Tkinter, YAML/JSON Config, PeopleSoft, Vercel

Languages: English (fluent), Arabic (fluent)